Exercice 1 :

1/ L'expression de H est donnée par : $H = -\frac{1}{2}\sum_{ij} J_{ij} S_i S_j - \sum_i B_i$

or $Z = \sum_{\{S\}} e^{-\beta H}$

$$\Rightarrow Z = \sum_{\{S\}} e^{\beta/2 \sum_{ij} J_{ij} S_i S_j + \beta \sum_i B_i S_i}$$

$$Z = \sum_{\{S\}} \prod_{i,j} e^{\beta/2 J_{ij} S_i S_j} \prod_{i=1}^{N} e^{\beta B_i S_i}$$

2/ $\langle S_i \rangle = \sum_{\{S\}} S_i \frac{e^{-\beta H}}{Z}$

$$= \frac{1}{Z} \sum_{\{S\}} S_i \, e^{\beta/2 \sum_{ij} J_{ij} S_i S_j - \beta \sum_i B_i S_i}$$

$$= \frac{1}{Z} \sum_{\{S\}} \frac{1}{\beta} \times \frac{\partial}{\partial B_i} e^{\beta/2 \sum_{ij} J_{ij} S_i S_j + \beta \sum_i B_i S_i}$$

$$= \frac{1}{\beta} \times \frac{1}{Z} \frac{\partial}{\partial B_i} \sum_{\{S\}} e^{\beta H}$$

$$= \frac{1}{\beta} \times \frac{1}{Z} \frac{\partial Z}{\partial B_i} = \frac{1}{\beta} \frac{\partial \ln Z}{\partial B_i} = K_B T \frac{\partial \ln Z}{\partial B_i}$$

$$= \frac{\partial (K_B T \ln Z)}{\partial B_i} = \frac{\partial F}{\partial B_i}$$

or $M = \sum \langle S_i \rangle \Rightarrow M = \sum_i -\frac{\partial F}{\partial B_i}$

3/ $\langle S_i S_j \rangle = \sum_{\{S\}} S_i S_j \frac{e^{\beta H}}{Z}$

$$= \frac{1}{Z} \sum_{\{S\}} S_i S_j \, e^{\beta/2 \sum_{ij} B_{ij} S_i S_j} e^{\beta \sum_i B_i S_i}$$

$$= \frac{1}{Z} \times \frac{1}{\beta} \frac{\partial}{\partial B_i} \sum_{\{S\}} S_j \, e^{\beta/2 \sum_{ij} B_{ij} S_i S_j} e^{\beta \sum_i B_i S_i}$$

$$= \frac{1}{Z} \times \frac{1}{\beta} \frac{\partial}{\partial B_i} \sum_{\{S\}} Z \times \frac{S_j e^{\beta H}}{Z} = \langle S_j \rangle$$

$$= \frac{1}{\beta} \times \frac{1}{\beta} \frac{\partial}{\partial B_i} Z \langle S_j \rangle$$

Scanné avec CamScanner

$$= \frac{1}{Z} \times \frac{1}{\beta} \frac{\partial}{\partial B_i} \left(\frac{1}{\beta} \times \frac{\partial \ln Z}{\partial B_j} \right)$$

$$= \frac{1}{\beta^2} \frac{1}{Z} \left[\frac{\partial Z}{\partial B_i} \frac{\partial \ln Z}{\partial B_j} + Z \frac{\partial^2 \ln Z}{\partial B_i \partial B_j} \right]$$

$$= \frac{1}{\beta^2} \left[\frac{\partial \ln Z}{\partial B_i} \frac{\partial \ln Z}{\partial B_j} + \frac{\partial^2 \ln Z}{\partial B_i \partial B_j} \right]$$

$$\Gamma_{ij} = \langle S_i S_j \rangle - \langle S_i \rangle \langle S_j \rangle$$

$$= \cancel{\frac{1}{\beta^2} \frac{\partial \ln Z}{\partial B_i} \frac{\partial \ln Z}{\partial B_j}} + \frac{1}{\beta} \frac{\partial^2 \ln Z}{\partial B_i B_j} \cancel{- \frac{1}{\beta^2} \frac{\partial \ln Z}{\partial B_i} \frac{\partial \ln Z}{\partial B_j}}$$

$$= \frac{1}{\beta^2} \frac{\partial^2 \ln Z}{\partial B_i \partial B_j} = \frac{1}{\beta^2} \frac{\partial}{\partial B_i \partial B_j} \left(\frac{-F}{k_B T} \right)$$

$$= -\frac{1}{\beta^2} \times \beta \frac{\partial F}{\partial B_i \partial B_j} = -\frac{1}{\beta} \frac{\partial^2 F}{\partial B_i \partial B_j}$$

$$\Gamma_{ij} = -\frac{1}{\beta} \frac{\partial^2 F}{\partial B_i \partial B_j}$$

$$\frac{\partial M}{\partial B_j} = \sum_i -\frac{\partial^2 F}{\partial B_i \partial B_j} = \sum_i -\beta \frac{\partial^2 F}{\partial B_i \partial B_j}$$

$$= \sum_j \beta \Gamma_{ij}$$

1) $B_i = B$ champ uniforme :

$$\sum_j \Gamma_{ij} = -\frac{1}{\beta} \frac{\partial^2 F}{\partial B} = \frac{1}{\beta} \frac{\partial M}{\partial B}$$

$$\chi = \beta \sum \Gamma_{ij}$$

$$\left[\langle \xi^2 \rangle - \langle \xi \rangle^2 \right] = \sum_i \beta \left[\langle \sum_i S_i^2 \rangle - \langle \sum_i S_i \rangle^2 \right]$$

$$= \beta \left[\sum_i \langle S_i^2 \rangle - \sum \langle S_i \rangle^2 \right]$$

Scanné avec CamScanner

$$\langle S_i^2 \rangle = \sum_{\{S\}} S_i^2 \frac{e^{\beta H}}{Z} = \frac{1}{Z} \sum S_i \frac{1}{\beta} \frac{\partial}{\partial B_i} e^{\beta H}$$

$$= \frac{1}{Z} \times \frac{1}{\beta} \times \frac{\partial}{\partial B_i} \sum \frac{S_i}{Z} \frac{e^{\beta H}}{Z}$$

$$= \frac{1}{Z} \cdot \frac{1}{\beta} \frac{\partial}{\partial B_i} \left(\frac{1}{Z} \cdot \langle S_i \rangle \right)$$

$$= \frac{1}{Z} \cdot \frac{1}{\beta} \frac{\partial}{\partial B_i} \left(\frac{1}{Z} \cdot \frac{1}{\beta} \frac{\partial \ln Z}{\partial B_i} \right)$$

$$= \frac{1}{Z} \frac{1}{\beta^2} \frac{\partial^2 Z}{\partial B_i} + \frac{1}{\beta^2} \frac{\partial^2 \ln Z}{\partial B_i^2}$$

$$= \frac{1}{\beta^2} \left(\frac{\partial \ln Z}{\partial B_i} \right)^2 + \frac{1}{\beta^2} \frac{\partial^2 \ln Z}{\partial B_i^2}$$

$$\langle S_i \rangle = \frac{1}{\beta} \left(\frac{\partial \ln Z}{\partial B_i} \right)$$

$$\Rightarrow \beta \left[\langle \varepsilon^2 \rangle - \langle \varepsilon \rangle^2 \right] = \sum \beta \left[\frac{1}{\beta^2} \left(\frac{\partial \ln Z}{\partial B_i} \right)^2 + \frac{1}{\beta^2} \frac{\partial^2 \ln Z}{\partial B_i^2} - \frac{1}{\beta^2} \left(\frac{\partial \ln Z}{\partial B_i} \right)^2 \right]$$

$$= \sum \frac{1}{\beta} \frac{\partial^2 \ln Z}{\partial B_i^2}$$

$$= \sum \frac{1}{\beta} \frac{\partial^2 \ln Z}{\partial B_i \partial B_i}$$

$$\Gamma_{ij} = \sum_{ij} \langle S_i S_j \rangle - \langle S_i \times S_j \rangle$$

$$= \langle \xi^2 \rangle - \langle \xi \rangle^2$$

$$= \beta \left[\sum \langle S_i^2 \rangle - \sum_i \langle S_i \rangle^2 \right]$$

$$= \beta \left[\sum_i \langle S_i \rangle - \sum_{ij} \langle S_i \times S_j \rangle \right]$$

$$= \beta \left[\sum_{ij} \langle S_i S_j \rangle - \sum_{ij} \langle S_i \rangle \langle S_j \rangle \right]$$

$$= \beta \sum_{ij} \Gamma_{ij}$$

: X

Scanné avec CamScanner

$$\langle S_i S_j \rangle = \sum S_i S_j \frac{e^{\beta/2 \sum_{ij} J_{ij} S_i S_j}}{Z} = Z_{th}$$

$$= \sum_{S_i} S_i S_j \frac{e^{\beta/2 \sum_{ij} J_{ij} S_i S_j}}{Z}$$

$$= \frac{2}{\beta} \cdot \frac{1}{Z} \frac{\partial}{\partial J_{ij}} \sum e^{\beta H}$$

$$= \frac{2}{\beta} \times \frac{1}{Z} \frac{\partial Z}{\partial J_{ij}} = \frac{2}{\beta} \frac{\partial \ln Z}{\partial J_{ij}}$$

Exercice (2)

$$M = \sum_i S_i \qquad \hat{H} = \hat{H}_0 - M \sum_i S_i$$

1/ la fluctuation de l'aimantation

$$(\Delta M)^2 = \langle (M - \langle M \rangle)^2 \rangle$$

$$= \langle (M^2 - 2M\langle M \rangle + \langle M \rangle^2 \rangle$$

$$= \langle M^2 \rangle - 2\langle M \rangle \langle M \rangle + \langle M \rangle^2$$

$$\boxed{(\Delta M)^2 = \langle M^2 \rangle - \langle M \rangle^2}$$

$$\langle M \rangle = \sum_{\{S\}} M \frac{e^{-\beta H}}{Z}$$

$$= \frac{1}{Z} \sum_{\{S\}} M e^{-\beta(H_0 - H \sum S_i)}$$

$$= \frac{1}{Z} \sum_{\{S\}} M e^{-\beta(H_0 - MH)}$$

$$= \frac{1}{Z} \sum_{\{S\}} \frac{1}{\beta} \frac{\partial}{\partial H} e^{-\beta(H_0 - MH)}$$

$$= \frac{1}{\beta} \cdot \frac{1}{Z} \frac{\partial}{\partial H} \sum_{\{S\}} e^{-\beta(H_0 - MH)}$$

$$= \frac{1}{\beta} \cdot \frac{1}{Z} \frac{\partial Z}{\partial H} = \frac{1}{\beta} \frac{\partial \ln Z}{\partial H} \Rightarrow \langle M \rangle^2 = \frac{1}{\beta^2} \left(\frac{\partial \ln Z}{\partial H} \right)^2$$

Scanné avec CamScanner

$$\langle S_i S_j \rangle = \sum_{\{S\}} S_i S_j \frac{e^{\beta/2 \sum_{ij} J_{ij} S_i S_j + \beta \sum_i h_i S_i}}{Z} = Z_{th}$$

$$= \sum_{\{S\}} S_i S_j \frac{e^{\beta/2 \sum_{ij} J_{ij} S_i S_j}}{Z}$$

$$= \frac{2}{\beta} \cdot \frac{1}{Z} \frac{\partial}{\partial J_{ij}} \sum e^{\beta H}$$

$$= \frac{2}{\beta} \cdot \frac{1}{Z} \frac{\partial Z}{\partial J_{ij}} = \frac{2}{\beta} \frac{\partial \ln Z}{\partial J_{ij}}$$

Exercice (2)

$$M = \sum_i S_i \qquad \hat{H} = \hat{H}_0 - M \sum_i S_i$$

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$$\boxed{(\Delta M)^2 = \langle M^2 \rangle - \langle M \rangle^2}$$

$$\langle M \rangle = \sum_{\{S\}} M \frac{e^{-\beta H}}{Z}$$

$$= \frac{1}{Z} \sum_{\{S\}} M e^{-\beta (H_0 - H \sum S_i)}$$

$$= \frac{1}{Z} \sum_{\{S\}} M e^{-\beta (H_0 - MH)}$$

$$= \frac{1}{Z} \sum_{\{S\}} 1/\beta \frac{\partial e^{-\beta (H_0 - MH)}}{\partial H}$$

$$= \frac{1}{\beta} \cdot \frac{1}{Z} \frac{\partial}{\partial H} \sum_{\{S\}} e^{-\beta (H_0 - MH)}$$

$$= \frac{1}{\beta} \cdot \frac{1}{Z} \frac{\partial Z}{\partial H} = \frac{1}{\beta} \frac{\partial \ln Z}{\partial H} \Rightarrow \langle M \rangle^2 = \frac{1}{\beta^2} \left(\frac{\partial \ln Z}{\partial H} \right)^2$$

□

Scanné avec CamScanner

$$\langle M^2 \rangle = \sum_{\{S_i\}} M^2 \frac{e^{-\beta H}}{Z}$$

$$= \frac{1}{Z} \sum_{\{S_i\}} M^2 e^{-\beta(H_0 - HM)}$$

$$= \frac{1}{Z} \times \frac{1}{\beta} \sum_{\{S_i\}} \frac{\partial}{\partial H} \left(e^{-\beta(H_0 - HM)} M \right)$$

$$= \frac{1}{\beta} \cdot \frac{1}{Z} \frac{\partial}{\partial H} \left(\sum_{\{S_i\}} e^{-\beta(H_0 - HM)} M \right)$$

$$= \frac{1}{\beta} \times \frac{1}{Z} \frac{\partial}{\partial H} \left(Z \cdot \frac{\sum e^{-\beta(H_0 - HM)}}{Z} M \right)$$

$$= \frac{1}{\beta} \cdot \frac{1}{Z} \frac{\partial}{\partial H} \left(Z \times \frac{1}{\beta} \frac{\partial \ln Z}{\partial H} \right)$$

$$= \frac{1}{\beta^2} \frac{1}{Z} \frac{\partial}{\partial H} \left(Z \frac{\partial \ln Z}{\partial H} \right)$$

$$= \frac{1}{\beta^2} \left(\frac{\partial \ln Z}{\partial H} \frac{\partial \ln Z}{\partial H} + \frac{\partial^2 \ln Z}{\partial H^2} \right)$$

$$= \frac{1}{\beta^2} \left(\left(\frac{\partial \ln Z}{\partial H} \right)^2 + \frac{\partial^2 \ln Z}{\partial H^2} \right)$$

$$\Rightarrow (\Delta M)^2 = \frac{1}{\beta^2} \left(\frac{\partial \ln Z}{\partial H} \right)^2 + \frac{1}{\beta^2} \frac{\partial^2 \ln Z}{\partial H^2} - \frac{1}{\beta^2} \left(\frac{\partial \ln Z}{\partial H} \right)^2$$

$$= \frac{1}{\beta^2} \frac{\partial^2 \ln Z}{\partial H^2}$$

$$\text{or} \quad \chi_T = \frac{\partial M}{\partial H} = \frac{\partial}{\partial H} \left(\frac{1}{\beta} \frac{\partial \ln Z}{\partial H} \right) = \frac{1}{\beta} \frac{\partial^2 \ln Z}{\partial H^2} = \beta (\Delta M)^2$$

$$= \frac{1}{K_B T} (\Delta M)^2$$

$$\Rightarrow \boxed{(\Delta M)^2 = K T \chi_T}$$

2)

$$\sum_{ij} \Gamma_{ij} = \sum_{ij} \langle S_i \rangle \langle S_j \rangle - \langle S_i \rangle \langle S_j \rangle$$

[illegible] par translation : $\langle S_i \rangle = \langle S_j \rangle = \langle S \rangle$

Scanné avec CamScanner

$$\langle \quad \rangle = \frac{1}{Z} \sum_{\{S\}} \quad e^{-\beta H} \qquad \langle \quad \rangle = \frac{Y_Z}{Z}$$

$$(\Delta H)^2 = \langle (H - \langle H \rangle)^2 \rangle = \langle H^2 \rangle - \langle H \rangle^2$$

$$\langle H \rangle = \sum_{\{S\}} H \frac{e^{-\beta H}}{Z} = \frac{1}{Z} \sum_{\{S\}} H e^{-\beta H}$$

$$= -\frac{1}{Z} \sum_{\{S\}} \frac{\partial e^{-\beta H}}{\partial \beta} = \frac{-1}{Z} \frac{\partial}{\partial \beta} \sum_{\{S\}} e^{-\beta H}$$

$$= \frac{-1}{Z} \frac{\partial Z}{\partial \beta} = -\frac{\partial \ln Z}{\partial \beta}$$

$$\langle H^2 \rangle = \sum_{\{S\}} H^2 \frac{e^{-\beta H}}{Z} = -\frac{1}{Z} \sum_{\{S\}} \frac{\partial}{\partial \beta} (H e^{-\beta H})$$

$$= -\frac{1}{Z} \frac{\partial}{\partial \beta} \sum_{\{S\}} H e^{-\beta H}$$

$$= -\frac{1}{Z} \frac{\partial}{\partial \beta} \left(Z \times \underbrace{\sum_{\{S\}} \frac{H e^{-\beta H}}{Z}}_{\langle H \rangle} \right)$$

$$= \frac{1}{Z} \frac{\partial}{\partial \beta} \left(Z \times \frac{\partial \ln Z}{\partial \beta} \right)$$

$$= \frac{1}{Z} \left[\frac{\partial Z}{\partial \beta} \times \frac{\partial \ln Z}{\partial \beta} + Z \times \frac{\partial^2 \ln Z}{\partial \beta^2} \right]$$

$$= \left(\frac{\partial \ln Z}{\partial \beta} \right)^2 + \frac{\partial^2 \ln Z}{\partial \beta^2}$$

$$\langle H^2 \rangle - \langle H \rangle^2 = \cancel{\left(\frac{\partial \ln Z}{\partial \beta} \right)^2} + \frac{\partial^2 \ln Z}{\partial \beta^2} - \cancel{\left(\frac{\partial \ln Z}{\partial \beta} \right)^2} = \frac{\partial^2 \ln Z}{\partial \beta^2}$$

$$C_v = \frac{\partial \langle H \rangle}{\partial T} = \frac{\partial}{\partial T} \left(-\frac{\partial}{\partial \beta} \ln Z \right) = \frac{\partial \beta}{\partial T} \times \frac{\partial}{\partial \beta} \left(-\frac{\partial}{\partial \beta} \ln Z \right)$$

$$= +\frac{1}{k_B T^2} \frac{\partial^2 \ln Z}{\partial \beta^2}$$

$$= \frac{1}{k_B T^2} (\Delta H)^2$$

Scanné avec CamScanner

$$\Rightarrow \boxed{(\Delta H)^2 = K_B T^2 C_V}$$

4/ $\Delta H = (K_B T^2 C_V)^{1/2} = K_B^{1/2} T C_V^{1/2}$

or il sait que $\langle H \rangle = \frac{3}{2} N K_B T$

$$\Rightarrow \frac{\partial \langle H \rangle}{\partial T} = \frac{3}{2} N K_B = C_V$$

En remplace dans (ΔH) : $\Delta H = K_B^{1/2} T \left(\frac{3}{2} N K_B\right)^{1/2}$

$$= \left(\frac{3}{2}\right)^{1/2} K_B T N^{1/2}$$

$$= \left(\frac{3}{2}\right)^{1/2} \times \frac{1}{\beta} N^{1/2}$$

$$\Delta H \propto N^{\frac{1}{2}}$$

Exercice 3

1/ a/ $t \to c$ $\quad f(t) = A t^{1/2} + B t^{1/4} + C t$

$$f(t) = t^{1/4} \left(A t^{1/2 - 1/4} + B + C t^{1 - 1/4} \right)$$

$$= t^{1/4} \left(A t^{1/4} + B + C t^{3/4} \right)$$

et on sait que : $\lambda = \lim_{t \to c} \frac{\ln |f(t)|}{\ln |t|}$

$$\Rightarrow \ln f(t) = \frac{1}{4} \ln t + \ln \left(A t^{1/4} + B + C t^{3/4} \right)$$

$$\Rightarrow \frac{\ln f(t)}{\ln t} = \frac{1}{4} + \frac{\ln \left(A t^{1/4} + B + C t^{3/4} \right)}{\ln t}$$

$$\lim_{t \to c} \frac{\ln f(t)}{\ln |t|} = \frac{1}{4} + \lim_{t \to 0} \frac{\ln \left| A t^{1/4} + B + C t^{3/4} \right|}{\ln |t|}$$

$$\Rightarrow \boxed{\lambda = \lim_{t \to 0} \frac{\ln f(t)}{\ln |t|} = \frac{1}{4}}$$

Scanné avec CamScanner

1) $f(t) = A t^{-1/3} (t+B)^{2/3}$

$\ln f(t) = \ln \left(A t^{-2/3} (t+B)^{2/3} \right)$

$= -\frac{2}{3} \ln At + \frac{2}{3} \ln (t+B)$

$= -\frac{2}{3} \ln A - \frac{2}{3} \ln t + \frac{2}{3} \ln (t+B)$

$\lim_{t \to 0} \frac{\ln f(t)}{\ln t} = -\frac{2}{3} \frac{\ln A}{\ln t} + \frac{2}{3} \frac{\ln (t+B)}{\ln t} \Rightarrow \boxed{\lambda = \frac{-2}{3}}$

2) $f(t) = A t^2 e^{-t}$

$\ln f(t) = \ln (A t^2 e^{-t}) = \ln A + 2 \ln t + \ln e^{-t}$

$= 2 \ln t + \ln A - t$

$\lambda = \lim_{t \to 0} \frac{\ln f(t)}{\ln (t)} = 2 + \frac{\ln A}{\ln t} - \frac{t}{\ln t} = 2.$

$\boxed{\lambda = 2}$

3) $f(t) = A t^2 e^{1/t}$

$\ln f(t) = \ln (A t^2 e^{1/t}) = \ln A + 2 \ln t + \frac{1}{t}$

$\lim_{t \to 0} \frac{\ln f(t)}{\ln (t)} = 2 + \frac{\ln (A e^{1/t})}{\ln t} \rightarrow$ indéfini

$f(t) = A \ln \left(\exp (1/t^4) - 1 \right)$

$t \to 0, \ t^4 \to 0 : \frac{1}{t^4} \to \infty$

$\exp \left(\frac{1}{t^4} \right) - 1 \sim \exp \left(\frac{1}{t^4} \right)$

$f(t) = A \ln \left(\exp \left(\frac{1}{t^4} \right) \right) = \frac{A}{t^4}$

$\Rightarrow \ln f(t) = \ln A - 4 \ln t \Rightarrow \lim_{t \to 0} \frac{\ln f(t)}{\ln t} = -4 + \frac{\ln A}{\ln t}$

Scanné avec CamScanner

a/ $f(t) = A \ln|t| + B$ on fait un changement de variable

$\Rightarrow$ $u = \ln t$

$t \to 0$

$u \to \infty$

$f(u) = Au + B$

$$\lambda = \lim_{u \to \infty} \frac{\ln(Au+B)}{u} = \lim_{u \to \infty} \frac{\ln\left(u\left(A + \frac{B}{u}\right)\right)}{u}$$

$$= \lim_{u \to \infty} \frac{\ln u}{u} + \frac{\ln\left(A + \frac{B}{u}\right)}{u} = 0$$

b/ $f(t) = A - Bt^{1/2}$ $\Rightarrow \ln(A - Bt^{1/2}) = \ln A$

$$\Rightarrow \lambda = \lim_{t \to 0} \frac{\ln A}{\ln t} = 0$$

c/ $f(t) = 1 \quad t < 0$

$f(t) = 2 \quad t > 0$

$$\lambda = \lim_{t \to 0} \frac{\ln f(t)}{\ln t} = 0$$

$$\lambda = \lim_{t \to 0} \frac{\ln 2}{\ln t} = 0$$

d/ $f(t) = A(t^2 + B^2)^{1/2} (\ln|t|)^2$

$$\ln f(t) = \ln A + \frac{1}{2}\ln(t^2 + B^2) + 2\ln(\ln t)$$

$$\lim_{t \to 0} \frac{\ln f(t)}{\ln t} = \lim_{t \to 0} \frac{\ln A}{\ln t} + \frac{\ln B}{\ln t} + \frac{2\ln(\ln t)}{\ln t}$$

$\frac{\ln A}{\ln t} \to 0$; $\frac{\ln B}{\ln t} \to 0$; $\frac{2\ln(\ln t)}{\ln t} \to 0$

on pose $u = \ln t$

$t \to 0$, $u \to \infty$

$$\Rightarrow \lim_{\substack{t \to 0 \\ u \to \infty}} \frac{2\ln u}{u} = 0 \quad \text{donc} \quad \lambda = 0$$

Scanné avec CamScanner

$f(H) = A\,t\,\mathrm{Ln}\,|H| + B$

$\mathrm{Ln}\,f(H) = \mathrm{Ln}\,(A\,t\,\mathrm{Ln}\,|H| + B)$

$\mathrm{Ln}\,t = u \Rightarrow t = e^u$

$$\frac{\mathrm{Ln}\,f(H)}{\mathrm{Ln}\,t} = \lim_{t\to 0} \frac{\mathrm{Ln}\,(A\,u\,e^u + B)}{u} = \lim_{t\to 0} \frac{\mathrm{Ln}\,B}{u} = 0$$

$\Rightarrow \lambda = 0$

Exercice 5

a) Mq : $f(x,y) = x^3 - y^2$??

on sait que $f(\lambda^a x, \lambda^b y) = \lambda f(x,y)$

$a = \frac{1}{3}$

$b = \frac{1}{2}$

$$f(x,y) = x^3 + y^2$$
$$= \lambda^{3a} x^3 + \lambda^{2b} y^2$$
$$= \lambda (x^3 + y^2)$$
$$= \lambda f(x,y)$$

$$f(x,y) = x^3 y^2 + x^2 y^3$$

$$f(\underbrace{\lambda^a x}_{x}, \underbrace{\lambda^b y}_{y}) = \lambda^{3a} x^3 \lambda^{2b} y^2 + \lambda^{2a} x^2 \lambda^{3b} y^3$$

$\Rightarrow$

$$\lambda^{3a}\lambda^{2b} = \lambda^{2a}\lambda^{3b}$$
$$\lambda^{3a+2b} = \lambda^{2a+3b} \Rightarrow 3a + 2b = 2a + 3b$$
$$\Rightarrow a = 1$$
$$b = 1$$

$$f(x,y) = x^{1/a} \hat{f}(x^{-b/a} y) = y^{1/b} \check{f}(y^{-a/b} x)$$

sachant que $f(\lambda^a x, \lambda^b y) = \lambda f(x,y)$

cas particulier $\lambda^a x = 1 \Rightarrow \lambda = \frac{1}{x^{1/a}}$

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Scanné avec CamScanner

$$f(x,y) = x^{1/a} f(1, \frac{y}{x^{b/a}})$$

$$= x^{1/a} f(1, y x^{-b/a})$$

$$\boxed{f(x,y) = x^{1/a} \hat{f}(1, y x^{-b/a})}$$

$$f(\lambda^a x, \lambda^b y) = \lambda f(x,y) \qquad \lambda^b y = 1 \Rightarrow \lambda =$$

$$f(x,y) = y^{1/b} f(\lambda^a x, 1)$$

$$= y^{1/b} \check{f}(\frac{x}{y^{a/b}}, 1)$$

$$\boxed{f(x,y) = y^{1/b} \check{f}(x y^{-a/b})}$$

3/

$$f(x,y) = x^3 + y^2$$

$$= x^3 \hat{f}(x^{-3/2} y) = y^2 \check{f}(y^{-2/3} x)$$

$$= x^3 (1 + \frac{y^2}{x^3}) = y^2 (1 + \frac{x^3}{y^2})$$

$$\hat{f}(x^{-3/2} y) = 1 + \frac{y^2}{x^3}$$

$$\check{f}(y^{-2/3} x) = 1 + \frac{x^3}{y^2}$$

Scanné avec CamScanner

Exercice (6)

$$H = aM(t + bM^2)^\theta$$

avec $1 < \theta < 2$ $\qquad t = \frac{T - T_c}{T_c}$

Calcule de γ :

$$\chi_T = \frac{\partial M}{\partial H} \quad \text{pour} \quad H \to 0 \quad t \to 0$$

Donc $\quad \frac{\partial H}{\partial M} \simeq a t^{\theta} \quad \chi_T = \frac{\partial M}{\partial H} = \frac{1}{a} t^{-\theta} \simeq |t|^{-\theta}$

$$\Rightarrow \quad \chi_T \simeq |t|^{-\theta} \quad \Rightarrow \quad \boxed{\gamma = \theta}$$

Calcule de δ :

$$H = aM(t + bM^2)^\theta$$

$$= aM\left(bM^2\left(\frac{t}{bM^2} + 1\right)\right)^\theta$$

$$= aM(bM^2)^\theta\left(\frac{t}{bM^2} + 1\right)^\theta$$

$$H = M^{1+2\theta} a b^\theta \left[1 + \theta \frac{t}{bM^2}\right]$$

$$H \sim M^{1+2\theta} \iff H \sim M^\delta$$

Donc $\quad \boxed{\delta = 1 + 2\theta}$

Calcule de β :

$$m \simeq (-t)^\beta$$

sur $H = 0$

$$\Rightarrow H = aM(t + bM^2)^\theta = 0$$

$$\Rightarrow t + bM^2 = 0$$

$$\Rightarrow M^2 = -\frac{t}{b} \Rightarrow M = \left(-\frac{t}{b}\right)^{1/2}$$

$$M \simeq (-t)^\beta \iff M = \left(-\frac{t}{b}\right)^{1/2}$$

$$\Rightarrow \boxed{\beta = \frac{1}{2}}$$

Scanné avec CamScanner

on prouve (1) $\gamma + 2\beta + \alpha \geq 2$

(2) $\alpha + \beta(1+\delta) \geq 2$

(1) − (2) $\Rightarrow \gamma + 2\beta + \alpha - \alpha - \beta(1+\delta) \geq 0$

$\Rightarrow \gamma + \beta - \beta\delta \geq 0$ pour $\begin{cases} \beta = 1/2 \\ \gamma = 0 \\ \delta = 1 + 2\varepsilon \end{cases}$

$\Rightarrow \varepsilon + \frac{1}{2} - \frac{1}{2}(1 + 2\varepsilon)$

$= \varepsilon + \frac{1}{2} - \frac{1}{2} - \varepsilon = 0$

les exposants critiques est bien vérifié.

Exercice 8

1/ on $G(P,r) = \dfrac{\psi(r, t^{(2-\alpha)/d})}{r^{d-2+\eta}} = r^{-(d-2+\eta)} \psi(r t^{(2-\alpha)/d})$ (1)

on sait que D'après le cours: $G(r,t) = r^{-\tau} e^{-r/\xi}$

or $\xi \sim |t|^{-\nu}$

$= r^{-(d-2+\eta)} e^{-r/t^{-\nu}}$

$= r^{-(d-2+\eta)} e^{-r t^{\nu}}$ (2)

(1) = (2) $\Rightarrow \dfrac{2-\alpha}{d} = \nu \Rightarrow$ $\boxed{d\nu = 2 - \alpha}$ Relation de Josephson

2/

Scanné avec CamScanner

Exo 9

$$H = -J \sum_{\langle i,j \rangle} S_i S_j$$

$$Z = \sum_{\{S\}} e^{-\beta H} = \sum_{\{S\}} e^{\beta J \sum_{\langle i,j \rangle} S_i S_j}$$

$$= \sum_{\{S\}} \prod_{\langle i,j \rangle} e^{\beta J S_i S_j} = \sum_{\{S\}} \prod_{\langle i,j \rangle} e^{L S_i S_j}$$

or $e^{L S_i S_j} = \mathrm{Ch}L + S_i S_j \mathrm{Sh}L$

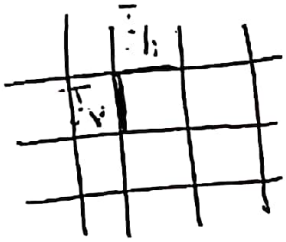
$$= (1 + S_i S_j \mathrm{Th}L)\,\mathrm{Ch}L$$

$\Rightarrow$ $Z = \sum_{\{S\}} \prod_{\langle i,j \rangle} \mathrm{Ch}L\,(1 + S_i S_j \mathrm{Th}L)$ on pose $\mathrm{Th}L = u$

$\Rightarrow$ $Z = \sum_{\{S\}} \prod_{\langle i,j \rangle} \mathrm{Ch}L\,(1 + S_i S_j u)$

$$= (\mathrm{Ch}L)^N \sum_{\{S\}} \left[1 + u \sum_{\langle i,j \rangle} S_i S_j + u^2 \sum_{\langle i,j \rangle \neq \langle k,l \rangle} (S_i S_j)(S_k S_l) \right]$$

$$= (\mathrm{Ch}L)^N\, 2^N \left[1 + \sum_n \Omega_n u^n \right]$$



1) Puisque on a un réseau Carré

$$H = -J_v \sum_{\langle i,j \rangle} S_i S_j - J_h \sum_{\langle k,l \rangle} S_k S_l$$

$$\Rightarrow Z = \sum_{\{S\}} e^{-\beta H} = \sum_{\{S\}} e^{-\beta\left(-J_v \sum_{\langle i,j \rangle} S_i S_j - J_h \sum_{\langle k,l \rangle} S_k S_l\right)}$$

$$= \sum_{\{S\}} e^{\beta J_v \sum_{\langle i,j \rangle} S_i S_j}\, e^{\beta J_h \sum_{\langle k,l \rangle} S_k S_l}$$

$L = \beta J_v$

$K = \beta J_h$

$$\Rightarrow Z = \sum_{\{S\}} e^{L \sum_{\langle i,j \rangle} S_i S_j}\, e^{K \sum_{\langle k,l \rangle} S_k S_l}$$

Scanné avec CamScanner

$$\begin{cases} e^{L S_i S_j} = \mathrm{Ch}L + S_i S_j \,\mathrm{Sh}L \\ e^{K S_k S_l} = \mathrm{Ch}K + S_k S_l \,\mathrm{Sh}K \end{cases} \qquad \mathrm{Th}K = u$$

$$= \sum_{\{S_i\}} \prod_{\langle ij \rangle} \mathrm{Ch}L \,(1 + S_i S_j u) \times \prod_{\langle kl \rangle} \mathrm{Ch}K \,(1 + S_k S_l u')$$

NO:

1/ on sait que le modèle d'Ising

$$H = -J \sum_{\langle ij \rangle} S_i S_j \qquad S_i = \pm 1$$

- le modèle de Potts s'écrit comme :

$$H = -J \sum_{\langle ij \rangle} \delta_{\sigma_i \sigma_j}$$

avec $\delta_{\sigma_i \sigma_j} = \dfrac{1 + \sigma_i \sigma_j}{2}$

$$\Rightarrow H = -J \sum_{\langle ij \rangle} \frac{1 + \sigma_i \sigma_j}{2}$$

$$= -\frac{J}{2} \sum_{\langle ij \rangle} \sigma_i \sigma_j + C$$

$$H = -J_{ising} \sum_{\langle ij \rangle} \sigma_i \sigma_j + C \qquad \text{pour } q = 2$$

avec $\begin{cases} \sigma_i = S_i \\ \sigma_j = S_j \end{cases} = 0$

$$\boxed{H = -J_{ising} \sum_{\langle ij \rangle} S_i S_j}$$

* pour $p = 2 \Rightarrow \delta_{ij} = \frac{1}{2}\left[\cos\left(\frac{2\pi}{p}(n_i - n_j)\right) + 1\right]$

$$H = -J_p \sum_{ij} \delta_{ij} = -J_p \frac{1}{2} \sum_{ij} \cos\left(\frac{2\pi}{p}(n_i - n_j)\right) + C$$

sachant que :

$$\cos\left(\frac{2\pi}{p}(n_i - n_j)\right) = \begin{cases} 1 & \text{si } n_i = n_j \\ -1 & \text{si } n_i - n_j = 1 \end{cases}$$

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$$H = -\frac{J_p}{2} \sum_{ij} \cos\left(\frac{2\pi}{p}(n_i - n_j)\right) + C$$

$$H = -J_c \sum_{ij} \cos\left(\frac{2\pi}{p}(n_i - n_j)\right) + C \Rightarrow \boxed{J_c = \frac{J_p}{2}}$$

→ Potts Model est équivalent à Clock Model.

* pour $p = 3$ on pose $S_{ij} = \frac{2}{3}\left[\cos\left(\frac{2\pi}{p}(n_i - n_j)\right) + \frac{1}{2}\right]$

$$\cos\left(\frac{2\pi}{p}(n_i - n_j)\right) = \begin{cases} 1 & \text{si } n_i = n_j \\ -\frac{1}{2} & \text{si } n_i - n_j = 1 \end{cases}$$

$$H = -J_p \sum_{ij} S_{ij} = -\frac{2}{3} J_p \sum_{ij} \cos\left(\frac{2\pi}{p}(n_i - n_j)\right) + C$$

$$= -J_c \sum_{ij} \cos\left(\frac{2\pi}{p}(n_i - n_j)\right) + C$$

→ Potts Model est équivalent à Clock Model $\Rightarrow \boxed{J_c = \frac{2}{3} J_p}$

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Série 13

1/ $H = -J \sum_{\langle ij \rangle} S_i S_j - H \sum_i S_i$ $\quad S_i = \pm 1$

l'énergie libre champ moyen. F_{CM}

$F_{CM} = \min_{\{\lambda_i\}} \{\ldots\}$. $H = H_0 + H_1$

$\Rightarrow H_1 = H - H_0$

$\Rightarrow$ l'inégalité de Bregelbov $F \leq F_0 + \langle H - H_0 \rangle_0$

$F \leq F_0\{\lambda\} + \langle H\{\lambda\} - H_0\{\lambda_i\} \rangle_0 = \phi(\lambda)$

Calcule F_0 : $F_0 = -K_B T \ln Z_0$

$H_0 = -\lambda \sum_i S_i$; champ effectif

$Z_0 = \sum_{\{S\}} e^{-\beta H_0} = \sum_{\{S\}} e^{\beta \lambda \sum_i S_i} = \sum_{\{S\}} \prod_{i=1}^{N} e^{\beta \lambda S_i}$

$= \sum_{\{S\}} \prod_{i=1}^{N} \left(Ch(\beta\lambda) + S_i \, Sh(\beta\lambda) \right)$

$= \left(Ch(\beta\lambda) \right)^N \sum_{\{S\}} \prod_{i=1}^{N} \left(1 + S_i \, Sh(\beta\lambda) \right)$

$= \left(Ch(\beta\lambda) \right)^N 2^N \Rightarrow Z_0 = \left(2\, Ch(\beta\lambda) \right)^N$

donc $\boxed{F_0 = -N K_B T \ln \left(2\, Ch(\beta\lambda) \right)}$

Calcul de $\langle H - H_0 \rangle_0$

$\langle H - H_0 \rangle = \langle -J \sum_{ij} S_i S_j - H \sum_i S_i - \lambda \sum_i S_i \rangle$

$= \langle -J \sum_{ij} S_i S_j - (H + \lambda) \sum_i S_i \rangle$

$= -J \sum_{ij} \langle S_i S_j \rangle - (H + \lambda) \sum_i \langle S_i \rangle$

1. On néglige les corrélation : $\Gamma_{ij} = \langle S_i S_j \rangle - \langle S_i \rangle \langle S_j \rangle = 0$

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$\langle S_i S_j \rangle = \langle S_i \rangle \langle S_j \rangle$

$\langle H - H_0 \rangle = -J \sum_{i,j} \langle S_i \rangle \langle S_j \rangle - (H + \lambda) \sum_i \langle S_i \rangle$

stem est invariante par translation : $\langle S_i \rangle_0 = \langle S_j \rangle_0 = \langle S \rangle_0$

$$\Rightarrow \langle H - H_0 \rangle = -J \sum_{i,j} \langle S \rangle_0^2 - (H + \lambda) \sum_i \langle S_i \rangle$$

$$\langle H - H_0 \rangle = -J \frac{zN}{2} \langle S \rangle_0^2 - (H + \lambda) N \langle S \rangle_0$$

ercices $\langle S \rangle_0$

$$\langle S \rangle_0 = \langle S_i \rangle_0 = \frac{\sum_{\{S_i\}} S_i \, e^{-\beta H_0}}{\sum_{\{S_i\}} e^{-\beta H_0}} = th(\beta\lambda)$$

$$\Rightarrow \boxed{\langle S \rangle_0 = th(\beta\lambda)}$$

$$\Rightarrow \boxed{\langle H - H_0 \rangle = -J \frac{zN}{2} th^2(\beta\lambda) - (H + \lambda) N th(\beta\lambda)}$$

$$\Rightarrow \lambda) = -NK_B T \ln\left(2ch(\beta\lambda)\right) - \frac{Jz N}{2} th^2(\beta\lambda) - (H - \lambda) N th(\beta\lambda)$$

$$\frac{d}{\lambda} = 0 \Rightarrow \lambda_{min} = ?$$

$$\vdots = -NK_B T \, 2\beta \frac{sh(\beta\lambda)}{2ch(\beta\lambda)} - \frac{Jz N}{2} \times 2 \cdot th(\beta\lambda) \cdot \left(th(\beta\lambda)\right)' + N th(\beta\lambda) - (H - \lambda) N \left(th(\beta\lambda)\right)' = 0$$

$$-N th(\beta\lambda) - Jz N th(\beta\lambda) \cdot \beta \frac{1}{ch^2(\beta\lambda)} + N th(\beta\lambda) - (H - \lambda) \frac{N\beta}{ch^2(\beta\lambda)} = 0$$

$$\frac{-N\beta}{ch^2(\beta\lambda)} \left\{ Jz \, th(\beta\lambda) + (H - \lambda) \right\} = 0$$

$$Jz \, th(\beta\lambda) + (H - \lambda) = 0$$

$$\boxed{\lambda_{min} = Jz \, th(\beta\lambda)_{min} + H}$$

λ_S

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aussi $\boxed{\lambda_{min} = \partial_\beta \langle s \rangle_0 + H}$

alors :

$$F_{CH} = -NK_B T \ln\left(2\cosh(\beta\lambda_{min})\right) - \frac{\partial_\beta N}{2}\tanh^2(\beta\lambda_{min})$$

$$- (H - \partial_\beta \tanh(\beta\lambda_{min}) - H)\, N \tanh(\beta\lambda_{min})$$

$$\boxed{F_{CM} = -NK_B T \ln\left(2\cosh(\beta\lambda_{min})\right) + \frac{\partial_\beta N}{2}\tanh^2(\beta\lambda_{min})}$$

or $\langle s \rangle_0 = \tanh(\beta\lambda_{min})$

$$= \tanh\left(\beta\left(\partial_\beta \tanh(\beta\lambda_{min}) + H\right)\right)$$

$$\boxed{\langle s \rangle_0 = \tanh\left(\beta\left(\partial_\beta \langle s_0 \rangle + H\right)\right)}$$ éq d'état

finalement :

$$\begin{cases} F_{CM} = -NK_B T \ln\left(2\cosh\left(\beta\left(\partial_\beta \langle s_0 \rangle + H\right)\right)\right) + \frac{\partial_\beta N}{2}\tanh^2(\beta\lambda_m \\ \langle s \rangle_0 = \tanh\left(\beta\left(\partial_\beta \langle s_0 \rangle + H\right)\right) \end{cases}$$

où $\langle s \rangle_0$ est la solution de l'équation d'état du modèle

$\tanh(\beta T_c \langle s \rangle) = 0$ pour $T \leq T_c$

$\Rightarrow \partial_\beta = k_B T_c$

2/ $\langle s \rangle_0 = \tanh(\beta\lambda_{min}) = \tanh\left(\beta\left(\partial_\beta \langle s_0 \rangle + H\right)\right)$

à $H = 0 \Rightarrow \langle s \rangle_0 = \tanh\left(\beta \partial_\beta \langle s \rangle_0\right)$

on utilise le développement de $\tanh(x) = x - \frac{x^3}{3}$

$$\langle s \rangle_0 = \tanh(\beta\partial_\beta \langle s \rangle_0) = \beta\partial_\beta \langle s \rangle_0 - \frac{(\beta\partial_\beta \langle s \rangle_0)^3}{3}$$

$$\Rightarrow \langle s \rangle_0 = \frac{T_c}{T}\langle s \rangle_0 - \frac{1}{3}\left(\frac{T_c}{T}\right)^3 \langle s \rangle_0^3$$

$$\Rightarrow \left(1 - \frac{T_c}{T}\right)\langle s \rangle_0 = -\frac{1}{3}\left(\frac{T_c}{T}\right)^3 \langle s \rangle_0^3$$

$$\Rightarrow \langle s \rangle_0^2 = -3\left(1 - \frac{T_c}{T}\right)\left(\frac{T}{T_c}\right)^3$$

Scanné avec CamScanner

$\langle s \rangle_c = -3\left(\frac{1}{T_c} - \frac{1}{T_c}\right)\left(\frac{1}{T_c}\right)^2$

$\langle s \rangle_0 \simeq -(-t)^{1/2} \sim (-t)^{\beta} \Rightarrow \boxed{\beta = \frac{1}{2}}$

$\langle M \rangle = th\left(\beta(H + J\mathfrak{z}\langle s \rangle)\right)$

$= th\left(\beta J\mathfrak{z}\langle s \rangle + \beta H\right)$

$= th\left(\beta J\mathfrak{z}\left(\langle s \rangle + \frac{H}{J\mathfrak{z}}\right)\right)$

$= th\left(\langle s \rangle + \frac{H}{J\mathfrak{z}}\right)$

à $T \sim T_c^-$

$J\mathfrak{z} = k_B T_c$

$\therefore \frac{H}{k_B T_c} = \frac{H}{J\mathfrak{z}} = t$

$\langle M \rangle = \langle s \rangle = \left(\langle s \rangle + \frac{H}{J\mathfrak{z}}\right) - \frac{1}{3}\left(\langle s \rangle + \frac{H}{J\mathfrak{z}}\right)^3$

$\Rightarrow \frac{H}{J\mathfrak{z}} = \frac{1}{3}\left(\langle s \rangle + \frac{H}{J\mathfrak{z}}\right)^3$

$\Rightarrow H = \frac{J\mathfrak{z}}{3}\left(\langle s \rangle + \frac{H}{J\mathfrak{z}}\right)^3$

$\Rightarrow H \sim \langle s \rangle^3 \Leftrightarrow H \sim \langle M \rangle^3 \Rightarrow \boxed{\delta = 3}$

!) on sait que $\chi_T = \frac{\partial \langle M \rangle}{\partial H} = \frac{\partial \langle s \rangle}{\partial H}$

avec $\langle s \rangle_0 = th\left(\beta(J\mathfrak{z}\langle s \rangle_0 + H)\right)$

$\frac{\partial \langle s \rangle}{\partial H} = \left(\beta + \beta J\mathfrak{z}\frac{\partial \langle s \rangle_0}{\partial H}\right) th'\left(\beta J\mathfrak{z}\langle s \rangle_0 + \beta H\right)$

$th'(x) = \frac{1}{Ch^2(x)} = 1 - th^2(x)$

$\frac{\partial \langle s \rangle}{\partial H} = \chi_T = \left(\beta + \beta J\mathfrak{z}\chi_T\right)\left[1 - th^2\left(\beta J\mathfrak{z}\langle s \rangle_0 + \beta H\right)\right]$

$\chi_T = \left(\beta + \beta J\mathfrak{z}\chi_T\right)\left(1 - \langle s \rangle_0^2\right)$

$\chi_T = \chi_T\left(\frac{\beta}{\chi_T} + \beta J\mathfrak{z}\right)\left(1 - \langle s \rangle_0^2\right)$

$1 = \frac{\beta}{\chi_T}\left(1 - \langle s \rangle_0^2\right) + \beta J\mathfrak{z}\left(1 - \langle s \rangle_0^2\right)$

Scanné avec CamScanner

$T < T_c \qquad (s \neq 0)$

$\chi = \dfrac{1 - \langle s \rangle^2}{J\beta(1 + \langle s \rangle^2)}$ $\qquad m = \langle s \rangle \sim (-t)^{\beta}$

donc $\beta = 1/2$

$= \dfrac{1 - t}{J\beta(1-t)} = \dfrac{2-t}{2t} = \dfrac{1}{2t} - \dfrac{1}{2}$

$\chi \sim \dfrac{1}{2t} \sim (t)^{-1} \Rightarrow \gamma = 1$

6/ on sait que $\tanh x = x - \dfrac{x^3}{3} + \dfrac{2x^5}{15} - \ldots$

$\langle s \rangle = \tanh(\beta J z \langle s \rangle + \beta H) = \tanh\left(\beta J z \left(\langle s \rangle + \dfrac{H}{Jz}\right)\right)$

pour $H = 0 \Rightarrow \langle s \rangle = \tanh(\beta J z \langle s \rangle)$

$\Rightarrow \langle s \rangle = \beta J z \langle s \rangle - \dfrac{1}{3}(\beta J z)^3 \langle s \rangle^3 + \dfrac{2}{15}(\beta J z)^5 \langle s \rangle^5$

avec $t = \dfrac{T - T_c}{T_c}$

$\Rightarrow \langle s \rangle = \left(\dfrac{T_c}{T}\right)\langle s \rangle - \dfrac{1}{3}\left(\dfrac{T_c}{T}\right)^3 \langle s \rangle^3 + \dfrac{2}{15}\left(\dfrac{T_c}{T}\right)^5 \langle s \rangle^5$

$= \dfrac{1}{t+1}\langle s \rangle - \dfrac{1}{3}\left(\dfrac{1}{t+1}\right)^3 \langle s \rangle^3 + \dfrac{2}{15}\left(\dfrac{1}{t+1}\right)^5 \langle s \rangle^5$

$\Rightarrow 1 - \dfrac{1}{t+1} = -\dfrac{1}{3}\left(\dfrac{1}{t+1}\right)^3 \langle s \rangle^2 + \dfrac{2}{15}\left(\dfrac{1}{t+1}\right)^5 \langle s \rangle^4$

$t \times \left(\dfrac{1}{t+1}\right) = -\dfrac{1}{3}\left(\dfrac{1}{t+1}\right)^3 \langle s \rangle^2 + \dfrac{2}{15}\left(\dfrac{1}{t+1}\right)^5 \langle s \rangle^4$

$\Rightarrow t = -\dfrac{1}{3}\left(\dfrac{1}{t+1}\right)^2 \langle s \rangle^2 + \dfrac{2}{15}\left(\dfrac{1}{t+1}\right)^4 \langle s \rangle^4$

$\Rightarrow \langle s \rangle^2 = \dfrac{t}{-\frac{1}{3}\left(\frac{1}{t+1}\right)^2} + \mathcal{E}(t^4)$

$\langle s \rangle^2 = -3t(1+t)^2$

$\Rightarrow \langle s \rangle = \sqrt{-3t}(1+2t)^{1/2}$

$\langle s \rangle^2 \simeq -3t(1+2t)$

$= \sqrt{-3t}\left(1 - 2 \times \dfrac{1}{2} t\right)$

$\simeq \sqrt{-3t}(1 - t)$

$\langle s \rangle \sim \sqrt{(-3t)(1+2t)}$

$\Rightarrow \boxed{\beta = 1}$

Scanné avec CamScanner

$$(1 - \langle s \rangle_0^2) = 1 - \beta J z (1 - \langle s \rangle_0^2)$$

$$\frac{\chi_T}{\beta (1 - \langle s \rangle_0^2)} = \frac{1}{1 - \beta J z (1 - \langle s \rangle_0^2)}$$

$$\Rightarrow \chi_T = \frac{\beta (1 - \langle s \rangle_0^2)}{1 - \beta J z (1 - \langle s \rangle_0^2)}$$

$$= \frac{\beta (1 - \langle s \rangle_0^2)}{\beta \left(\frac{1}{\beta} - J z (1 - \langle s \rangle_0^2) \right)}$$

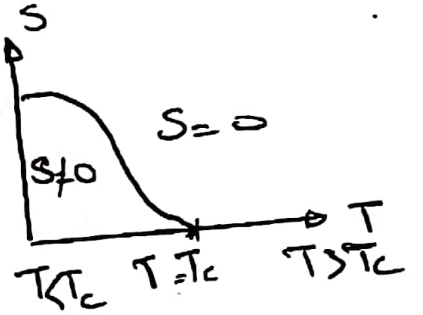
$$= \frac{1 - \langle s \rangle_0^2}{k_B T - J z (1 - \langle s \rangle_0^2)}$$

$$= \frac{1 - \langle s \rangle_0^2}{J z \left(\frac{k_B T}{J z} - 1 + \langle s \rangle_0^2 \right)}$$

$$= \frac{1 - \langle s \rangle_0^2}{J z \left(\frac{T}{T_c} - 1 + \langle s \rangle_0^2 \right)} = \frac{1 - \langle s \rangle_0^2}{J z (t + \langle s \rangle_0^2)}$$

($t = \frac{T}{T_c} - 1$)

/ $$\chi_T = \frac{1 - \langle s \rangle_0^2}{J z (t + \langle s \rangle_0^2)}$$



pour $T > T_c \rightarrow s$

$$\Rightarrow \chi_T \simeq \frac{1}{J z t} = \frac{1}{J z} t^{-1} \sim |t|^{-1}$$

or $\chi_i \sim |t|^{-\gamma} \Rightarrow \gamma = 1$

Scanné avec CamScanner

Exercice 14

1/ $F_{Cm} = -NK_BT \ln\left|2\cosh(\beta\lambda_{min})\right| + \frac{N\varepsilon_0}{2}\tanh^2(\beta\lambda_{min})$

$= -NK_BT \ln\left|2\cosh\left(\beta\varepsilon_0 z\langle S\rangle + \beta H\right)\right| + \frac{N\varepsilon_0}{2}\tanh^2\left(\beta\varepsilon_0 z\langle S\rangle + \beta H\right)$

pour $H = 0$

$\Rightarrow F_{Cm} = -NK_BT \ln\left|2\cosh\left(\beta\varepsilon_0 z\langle S\rangle\right)\right| + \frac{N\varepsilon_0}{2}\tanh^2\left(\beta\varepsilon_0 z\langle S\rangle\right)$

$F_{Cm} = -NK_BT \ln\left|2\cosh\left(\beta\varepsilon_0 z\langle S\rangle\right)\right| + \frac{N\varepsilon_0}{2}\langle S\rangle^2$

$F_{Cm} = -NK_BT \ln 2 - NK_BT \ln\left|\cosh\left(\beta\varepsilon_0 z\langle S\rangle\right)\right| + \frac{N\varepsilon_0}{2}\langle S\rangle^2$

$= -NK_BT \ln 2 - NK_BT \ln\left(1 + \frac{(\beta\varepsilon_0 z)^2}{2}\langle S\rangle^2\right) + \frac{N\varepsilon_0}{2}\langle S\rangle^2$

$= -NK_BT \ln 2 - NK_BT\left(\frac{(\beta\varepsilon_0 z)^2}{2}\langle S\rangle^2 - \frac{(\beta\varepsilon_0 z)^4}{8}\langle S\rangle^4\right) + \frac{N\varepsilon_0}{2}\langle S\rangle^2$

$= -NK_BT \ln 2 - NK_BT\frac{(\beta\varepsilon_0 z)^2}{2}\langle S\rangle^2 - \frac{NK_BT}{8}(\beta\varepsilon_0 z)^4\langle S\rangle^4 + \frac{N\varepsilon_0}{2}\langle S\rangle^2$

$= -NK_BT \ln 2 - \frac{NK_BT}{2}\left(\frac{T_c}{T}\right)^2\langle S\rangle^2 + \frac{NK_BT}{8}\left(\frac{T_c}{T}\right)^4\langle S\rangle^4 + \frac{N\varepsilon_0}{2}\langle S\rangle^2$

$= -NK_BT \ln 2 - \frac{NK_BT}{2}\left(\frac{1}{1+t}\right)^2(3t)(1+t)^2 + \frac{NK_BT}{8}\left(\frac{1}{1+t}\right)^4(3t)^2(1+t)^4 + \frac{N\varepsilon_0}{2}(3t)(1+t)^2$

$= -NK_BT \ln 2 - \frac{3}{2}NK_BT(t) + \frac{NK_BT}{2}(3t)^2 + \frac{N\varepsilon_0}{2}(3t)(1+t)^2$

2/ . pour $T > T_c$ $\quad S = 0$

$\Rightarrow F_{Cm} = -NK_BT \ln 2 \Rightarrow C_v = -T\left(\frac{\partial^2 F}{\partial T^2}\right) = 0$

$C_v = 0$ pour $T > T_c$.

Scanné avec CamScanner

$\mathcal{H} = -J \sum_{\langle ij \rangle} \delta_i \delta_j - H \sum_i \delta_i \quad \delta_i = \pm 1$

Exercice 13

$\mathcal{H}_0 = -\lambda \sum \delta_i \qquad \mathcal{H} = \mathcal{H}_0(\lambda) + \mathcal{H}_1(\lambda)$

H

$\mathcal{H}_1(\lambda) = \mathcal{H} - \mathcal{H}_0(\lambda) = -J \sum_{\langle ij \rangle} \delta_i \delta_j - (H - \lambda) \sum_i \delta_i$

$\langle \mathcal{H}_1 \rangle_0 = \langle \mathcal{H} - \mathcal{H}_0 \rangle_0 = -J \sum_{\langle ij \rangle} \langle \delta_i \rangle \langle \delta_j \rangle_0 - (H - \lambda) \sum_i \langle \delta_i \rangle_0$

invariante par translation : $\langle \delta_i \rangle_0 = \langle \delta_j \rangle_0 = \langle \delta \rangle_0$

$\Rightarrow \quad \langle \mathcal{H}_1 \rangle_0 = -J \left(\sum_{\langle ij \rangle} 1 \right) \langle \delta \rangle_0^2 - (H - \lambda) \left(\sum_i 1 \right) \langle \delta \rangle_0$

$= -J \frac{zN}{2} \langle \delta \rangle_0^2 - (H - \lambda) N \langle \delta \rangle_0$

$F(\lambda) = F_0(\lambda) + \langle \mathcal{H}_1 \rangle_0 \qquad \langle \delta \rangle_0 = th(\beta \lambda)$

$\mathcal{Z}_0 = [2 ch(\beta \lambda)]^N \qquad F_0(\lambda) = -N k_B T \ln [2 ch(\beta \lambda)]$

$F(\lambda) = \Phi(\lambda) = -N k_B T \ln [2 ch(\beta \lambda)] - \frac{JzN}{2} th^2(\beta \lambda) - (H - \lambda) N th(\beta \lambda)$

$\left. \frac{\partial F}{\partial \lambda} \right|_{\lambda = \lambda^* = \lambda_{min}} = 0$

$\Leftrightarrow \quad -N \cancel{th(\beta \lambda^*)} - \frac{JzN}{2} \cdot 2\beta \frac{th(\beta \lambda)^*}{ch^2(\beta \lambda^*)} + N \cancel{th(\beta \lambda^*)} - (H - \lambda^*) N \frac{\beta}{ch^2(\beta \lambda^*)} = 0$

$\Rightarrow \quad -\frac{N}{ch^2(\beta \lambda^*)} \left\{ \beta J z \, th(\beta \lambda^*) + \beta (H - \lambda^*) \right\} = 0$

$\Rightarrow \quad \beta J z \, th(\beta \lambda^*) + \beta (H - \lambda^*) = 0$

$\Rightarrow \quad \boxed{J z \, th(\beta \lambda^*) = \lambda^* - H} \qquad th(\beta \lambda^*) = \frac{\lambda^* - H}{J z}$

$F_{cm} = \Phi(\lambda^*) = -N k_B T \ln [2 ch(\beta \lambda^*)] - \frac{JzN}{2} \frac{(\lambda^* - H)^2}{(Jz)^2} + N \frac{(\lambda^* - H)^2}{Jz}$

$= -N k_B T \ln 2 ch(\beta \lambda^*) + \frac{N}{2Jz} (\lambda^* - H)^2$

$m = \langle \delta \rangle_0 = \frac{-1}{N} \frac{\partial F_{cm}}{\partial H} = \frac{1}{Jz} (\lambda^* - H) \Rightarrow \boxed{Jz \langle \delta \rangle_0 + H = \lambda^*}$

n : nombre de plus proches voisins

M, N : ...

donc : $\boxed{m = \langle \delta \rangle_0 = th [\beta (Jz \langle \delta \rangle_0 + H)]}$

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$T < T_c$ $\quad H = 0$ $\quad m = \text{th}\,\beta J z m$

$$\sim \beta J z m - \frac{1}{3}(\beta J z)^3 m^3$$

$$\Rightarrow 1 = \beta J z - \frac{1}{3}(\beta J z)^3 m^2$$

$$\frac{1}{3}(\beta J z)^3 m^2 = \beta J z - 1$$

on pose : $y = \text{th}(\beta J z m) \simeq \beta_c J z m = m$

à $T = T_c \Rightarrow \beta_c J z = 1 \Rightarrow \boxed{J z = k_B T_c}$

$$\Rightarrow m^2 \frac{1}{3}\left(\frac{\beta}{\beta_c}\right)^3 = \frac{\beta}{\beta_c} - 1 = \frac{T_c}{T} - 1 = \frac{T_c - T}{T}$$

? l'ordre le plus bas $\Rightarrow m^2 \propto T_c - T \Rightarrow m \propto (T_c - T)^{1/2}$

$$\Rightarrow \boxed{\beta = 1/2}$$

3°/ $T = T_c$ $\quad m = \text{th}(m + \beta_c H) \approx m + \beta_c H - \frac{1}{3}(m + \beta_c H)^3$

$$m = \text{th}(m + \beta_c H) \sim m + \beta_c H - \frac{1}{3}(m + \beta_c H)^3$$

$$\Rightarrow H \sim (m + \beta_c H)^3$$

$$\sim m^3 + \cdots$$

$$\Rightarrow \boxed{\delta = 3}$$

4°/ $\quad m = \text{th}\,\beta(J z m + H) = \text{th}\left(\frac{J z}{k_B T} m + \frac{H}{k_B T}\right)$

$m = \langle S \rangle_0$ $\quad T = T_c \Rightarrow \beta_c J z = 1 \Rightarrow J z = k_B T_c$

$$\frac{J z}{k_B T} = \frac{T_c}{T} = \frac{1}{1+t} \qquad t = \frac{T - T_c}{T_c}$$

$$\Rightarrow m = \text{th}\left(\frac{1}{1+t} m + \frac{1}{J z} \cdot \frac{H}{1+t}\right)$$

$$m = \text{th}\left(\frac{1}{1+t} m + \frac{1}{J z} \cdot \frac{H}{1+t}\right)$$

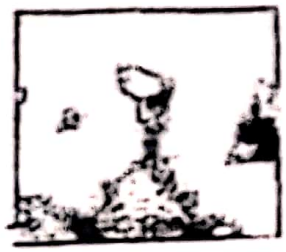
$\text{th}(x) \simeq x - \frac{x^3}{3}$ $\quad x = \frac{1}{(1+t)} m + \frac{1}{J z} \cdot \frac{H}{(1+t)}$

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$\frac{\partial}{\partial H} \cdot \frac{\partial}{\partial H} - x \cdot \frac{\partial}{\partial H} = (1-x^2) \frac{\partial x}{\partial H}$

avec $\frac{\partial x}{\partial H} = \frac{1}{(1+t)} \left(\frac{\partial m}{\partial H} + \frac{1}{Jz} \right)$

$\lim_{H \to 0} (\quad) = \frac{(1+t)^2 - m^2}{(1+t)^2}$



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$\chi = \frac{\partial m}{\partial H}\Big|_{H \to 0} = \frac{(1+t)^2 - m^2}{(1+t)^3} \left(\chi + \frac{1}{Jz} \right)$

donc on trouve (after the temperature) $\chi = \frac{1}{Jz} \left[\frac{(1+t)^2 - m^2}{t + m^2} \right]$

$1 \gg t \Rightarrow \boxed{\chi = \frac{1 - m^2}{Jz(t+m^2)}}$ ou $\chi = \frac{1 - \langle s \rangle_0^2}{Jz(t + \langle s \rangle_0^2)}$

5% . à $t > 0$, $m = 0$ (paramagnétique)

$\chi \simeq \frac{1}{Jzt} \sim t^{-1}$; $\chi \sim |t|^{-\gamma} \Rightarrow \boxed{\gamma = 1}$

. à $t < 0$ (ferromagnétique)

$m^2 = \langle s \rangle_0^2 = -3t \Rightarrow \chi \simeq \frac{1+3t}{Jz(-2t)} \sim \frac{-1}{2Jzt} \sim \dots t^{-1}$

$\chi \sim |t|^{-\gamma}$

l'exposant critique $\boxed{\gamma = 1}$

$\Rightarrow$

6% $\langle s \rangle_0 = \text{th}(\beta J z \langle s \rangle_0) = \text{th}\left(\frac{1}{t+1} \langle s \rangle_0 \right)$

$H = 0$

$= \frac{\langle s \rangle_0}{t+1} - \frac{\langle s \rangle_0^3}{3(1+t)^3} + O\left(\frac{\langle s \rangle_0^5}{(1+t)^5} \right)$

$= \langle s \rangle_0 (1-t) - \langle s \rangle_0^3 / 3 + O\left(\langle s \rangle_0 t^2, \langle s \rangle_0^3 t, \langle s \rangle_0^5 \right)$

$\Rightarrow \quad -t = \langle s \rangle_0^2 / 3 + O\left(t^2, \langle s \rangle_0^2 t, \langle s \rangle_0^4 \right)$

$\Rightarrow \quad -t \approx \langle s \rangle_0^2 / 3 \quad \Rightarrow \quad \langle s \rangle_0 \approx (-3t)^{1/2}$

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$$\chi = \frac{(1+t)^2 - m^2}{(1+t)^3}\chi + \frac{1}{\sqrt{3}}\frac{(1+t)^2 - m^2}{(1+t)^3}$$

$$\Rightarrow \left\{1 - \frac{(1+t)^2 - m^2}{(1+t)^3}\right\}\chi = \frac{1}{\sqrt{3}}\left\{\frac{(1+t)^2 - m^2}{(1+t)^3}\right\}$$

$$\Rightarrow \chi = \frac{(1+t)^3 - (1+t)^2 - m^2}{\sqrt{3}\left[(1+t)^2 - m^2\right]} \qquad \langle s\rangle_0 = -3t^{1/2}(1+\Omega t)$$

$$= \frac{1}{\sqrt{3}}\left\{\frac{(1+t)^3 - (1+t)^2 - m^2}{(1+t)^2 - m^2}\right\}$$

$$(1+t)^3\chi = \chi\left[(1+t)^2 - m^2\right] + \frac{1}{\sqrt{3}}\left((1+t)^2 - m^2\right) \qquad t = -\frac{\langle s\rangle_0^2}{3}$$

$$\Rightarrow \chi = \frac{}{\sqrt{3}\left((1+t)^2 - m^2\right)}$$

$$\left((1+t)^3 - (1+t)^2 + m^2\right)\chi = \frac{(1+t)^2 - m^2}{\sqrt{3}}$$

$$\Rightarrow \chi = \frac{(1+t)^2 - m^2}{\sqrt{3}(t+m^2)} = \frac{1-m^2}{\sqrt{3}(t+m^2)}$$

$$\tanh x \underset{0}{\simeq} x - \frac{x^3}{3} + \frac{2}{15}x^5 + O(x^7) \qquad (\text{"th 'odd function of } x\text{"})$$

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